

Supplementary Information

Antarctic sea ice retreated into pre-existing waves: swell attenuation, not fetch, controls wave exposure at the ice edge

S1. Sub-period Granger causality stability

To test whether the Granger causality results are robust to the 2016 regime shift (which violates the stationarity assumption), we repeated all six tests separately for the pre-2016 (1979–2015, $n = 444$ months) and post-2016 (2016–2024, $n = 108$ months) periods. Results are shown in Table S1.

Table S1: Sub-period Granger causality tests. p_{Bonf} corrected for 6 tests within each period.

Link	Pre-2016 ($n = 444$)			Post-2016 ($n = 108$)		
	F	p	p_{Bonf}	F	p	p_{Bonf}
SIC $\rightarrow$ Fetch	15.98	<0.001	<0.001	21.08	<0.001	<0.001
Fetch $\rightarrow$ SWH	0.28	0.756	1.000	3.60	0.061	0.364
SWH $\rightarrow$ Ice edge	2.57	0.038	0.226	1.18	0.280	1.000
Ice edge $\rightarrow$ SIC	2.03	0.109	0.652	1.37	0.258	1.000
Wind $\rightarrow$ SWH	2.88	0.057	0.342	3.73	0.056	0.337
SIC $\rightarrow$ SWH	2.71	0.068	0.406	5.80	0.018	0.107

The SIC $\rightarrow$ Fetch link is the only relationship that survives Bonferroni correction in both sub-periods. The post-2016 SIC $\rightarrow$ SWH link ($p = 0.018$) is stronger than the pre-2016 counterpart ($p = 0.068$) but remains marginal after Bonferroni correction ($p_{\text{Bonf}} = 0.107$) due to the small post-2016 sample size ($n = 108$).

S2. ERA5 wave model masking sensitivity

The ERA5 WAM does not compute wave fields in grid cells where SIC exceeds 30%. To test whether the SIC $\rightarrow$ SWH statistical link is an artifact of this masking, we classified Southern Ocean grid points (50–70°S) into two categories:

- **Always ice-free:** SIC < 15% in > 95% of all months ($n = 16,508$ grid points)
- **Transitional:** SIC crosses the 30% threshold in 10–90% of months ($n = 10,886$ grid points)

If the SIC $\rightarrow$ SWH link were a masking artifact, transitional grid points should show stronger SWH trends (as they gain wave data when SIC drops below 30%). Instead, always-ice-free grid points show a *stronger* SWH trend ($+0.046$ m decade $^{-1}$, $p = 0.0002$) than transitional points ($+0.019$ m decade $^{-1}$, $p = 0.056$). The SIC $\rightarrow$ SWH Granger link is not significant at either type of fixed grid point (always-ice-free: $p = 0.40$; transitional: $p = 0.97$), confirming that the signal arises from ice-edge-relative sampling rather than grid-level masking.

S3. Wave energy flux calculation

Wave energy flux per unit crest length is estimated as:

$$P = \frac{\rho g^2}{64\pi} H_s^2 T_p$$

where $\rho = 1025 \text{ kg m}^{-3}$ is seawater density, $g = 9.81 \text{ m s}^{-2}$, H_s is significant wave height, and T_p is peak wave period (approximated by MWP). At the ice edge, pre-2016 values ($H_s = 3.24 \text{ m}$, $T_p = 8.97 \text{ s}$) yield $P = 46.1 \text{ kW m}^{-1}$; post-2016 values ($H_s = 3.27 \text{ m}$, $T_p = 8.97 \text{ s}$) yield $P = 47.2 \text{ kW m}^{-1}$. The +2.4% increase is driven entirely by the small SWH increase; MWP is unchanged. In winter (JJA), the energy flux change is +0.1% (66.6 to 66.7 kW m^{-1}), confirming that the wave energy field at the ice edge is effectively stationary.

S4. Partial Granger causality formulation

The partial Granger test for “SIC $\rightarrow$ SWH | Wind” compares two nested models:

Reduced model (no SIC lags):

$$\text{SWH}_t = \alpha_0 + \sum_{l=1}^L \beta_l \text{SWH}_{t-l} + \sum_{l=1}^L \gamma_l \text{Wind}_{t-l} + \epsilon_t$$

Full model (includes SIC lags):

$$\text{SWH}_t = \alpha_0 + \sum_{l=1}^L \beta_l \text{SWH}_{t-l} + \sum_{l=1}^L \gamma_l \text{Wind}_{t-l} + \sum_{l=1}^L \delta_l \text{SIC}_{t-l} + \epsilon_t$$

The F -test compares the residual sum of squares:

$$F = \frac{(\text{RSS}_{\text{red}} - \text{RSS}_{\text{full}})/L}{\text{RSS}_{\text{full}}/(T - 2L - L - 1)}$$

On annual data, $F = 1.97$, $p = 0.14$ (SIC effect absorbed by wind collinearity). In JJA only, $F = 2.84$, $p = 0.040$ (SIC retains independent predictive power).

References